Experiment 2: The Effect of Chunking on Visual Memory Recall

1. Core Subject

This experiment investigates the **Chunking Principle** in Human-Computer Interaction, which suggests that users can retain more information when it is organized into small, manageable groups rather than presented as a massive, unstructured data set.

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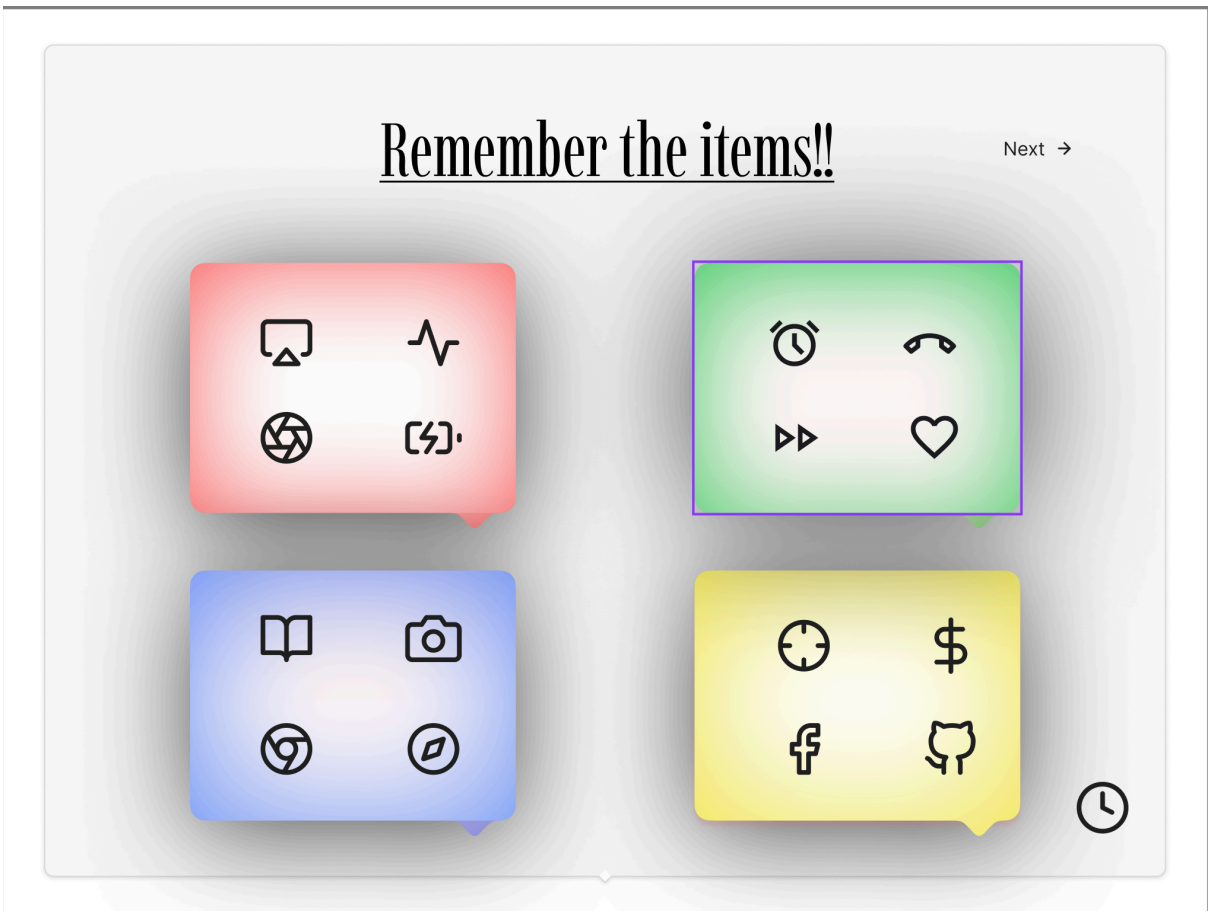
2. Experiment Methodology

The experiment utilized a "Memory Recall Task" designed in Figma to test user retention across structured and unstructured screens.

- **Task 1 (Memory Phase):** Users are presented with 16 icons organized into four distinct, color-coded "chunks" (Red, Green, Blue, Yellow) for a duration of 5 seconds.



- **Task 2 (Recall Phase):** Users are then shown a randomized, unstructured grid of icons and asked to select the ones they remember.

- **Measurement:** Success is measured by the number of items correctly recalled versus the total number of items presented.

3. Design Principles Applied

Principle	Implementation in Experiment
Miller's Law	The design organizes items into small groups (4 per box) to avoid overwhelming the user's short-term memory capacity (7 ± 2 items).
Visual Encoding	Different background colors for each chunk (Red, Green, Blue, Yellow) provide a secondary visual cue to help users categorize and retrieve information.
Recognition vs. Recall	The final step transitions from pure recall to recognition by providing a selection pool, demonstrating how UI can aid cognitive retrieval.

4. Observations and Findings

Optimized Interface (Chunking)

The "Remember the items!!" screen utilizes **Proximity** and **Similarity** (from Gestalt Principles) to create visual "anchors". By grouping the icons into 2×2 clusters within color-coded boxes, the cognitive load is significantly reduced.

The Recall Barrier

Even with optimized chunking, the "Select the items" screen introduces **Visual Noise** by placing the target icons within a sea of "distractor" icons in an unorganized layout. This transition highlights how easily memory is disrupted when the structured "chunks" are removed.

5. Conclusion

The experiment demonstrates that **Chunking** is essential for high-retention UI design. Without structured organization, users struggle to filter important information from noise. Designing for short-term memory involves creating clear visual boundaries and grouped categories to facilitate faster processing and better data retrieval.